BRAC University | CSE-460: VLSI Design | Summer 25 | Faculty: UPM

Assignment - 02 | Total Marks - 60

Q1. [CO2] - Marks 20

Design a FSM to detect the sequence "1001" using the binary encoding scheme:

(a) Considering a Moore type FSM **[10]**
(b) Considering a Mealy type FSM **[10]**

(You need to draw the State diagram, Construct State Table and State Assigned Table, Find the expressions for Next States and Output)

Q2. [CO2] - Marks 10

The input-output behavior of a sequence detector FSM is given below.																			
CLK	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	t_9	t_{10}	t_{11}	t_{12}	t_{13}	t_{14}	t_{15}	t_{16}	t_{17}	t_{18}	
w	**1**	**0**	**0**	**1**	1	**1**	**0**	0	0	0	0	0	0	**0**	**1**	**0**	0	0	
z	0	**1**	0	**1**	0	0	**1**	0	0	0	0	0	0	0	**1**	**1**	0	0	

(a)	**Identify** the <u>sequence</u> and the <u>type</u> of FSM for the above input-output combinations. [<u>Hint:</u> The above table contains two sequences]	**1+1**
(b)	**Design** the state diagram, the state table, and the ***gray-encoded*** state assigned table. **[Overlapping allowed].**	**3+2+2**
(c)	**Determine** the logic expressions of the next state and output variables using k-maps.	**6**

Q3. [CO2] - Marks 10

You are tasked with designing a CD/DVD drive's control circuitry using a Moore-type finite-state machine. The drive has one input push button w, which generates w = 1when pushed, and w = 0 when un-pushed. A w = 1 input causes the tray to open if it was already closed, or causes the tray to close if it was already open. The control circuit has only one output signal, z, which determines if the CD/DVD drive should be open (z = 1) or closed (z = 0) with the following instructions:

● If the tray is closed, pushing the button (setting w = 1) should generate the signal (z = 1) to open the tray
● If the tray has been open for less than 3 clock cycles, pushing the button (setting w = 1) should generate the signal (z = 0) to close the tray
● The tray should automatically close after being open for 3 clock cycles
● For simplicity, assume that the time required for opening and closing the tray is negligible compared to the system clock's time period

(a) Draw the state diagram for the machine.
(b) Derive a state-assigned table from (b) using the binary-encoding technique.

(c) Find the next-state and output logic expressions using k-map.

Q4. [CO2] - Marks 10

Consider a sequential circuit with two inputs (***u1*** & ***u2***) and a single output, ***z***.
CLEO is a prestigious conference where researchers from all over the world submit their research articles. These articles are handled by an **Editor** and a **Reviewer**. They review each article *independently,* and either approve a submitted article (denoted by 1) or reject it (denoted by 0). The decisions of the **Editor** are given by a bit stream, ***u1***, whereas the decisions of the **Reviewer** are given by ***u2***. **If for three consecutive articles, both of their decisions are the same (i.e. *u1* = *u2* for three consecutive clock cycles), then they receive a consistency reward (z = 1) from the conference committee *after* a period.**
Here, the decisions of the Editor & the Reviewer for a number of submitted articles & their corresponding reward instances are given below:

u1	0	1	1	0	1	1	1	0	0	0	1
u2	1	1	1	0	0	0	1	0	0	1	1
z	0	0	0	0	1	0	0	0	0	1	0

(a)	What type of FSM would you need for the given problem?	**1**
(b)	**Design** the state diagram for the corresponding FSM.	**4**
(c)	**Derive** the state-assigned table from your state diagram in (a).	**4**
(d)	**Evaluate** the <u>next state</u> & <u>output</u> logic expressions to implement the FSM & **calculate** the number of flip-flops required for the circuit implementation. (You don't need to draw the circuit implementation.)	**5 + 1**

Q5. [CO1] - Marks 10

Design a finite state machine (FSM) for a car alarm system. The system has two binary inputs: x, indicating the lock status of the car door (**x = 1 if unlocked, x = 0 if locked**) and y, indicating the door status (**y = 1 if the door is open, y = 0 if closed**). The output z is used to control the alarm: z = 1 when the alarm is active, and z = 0 otherwise. The alarm should be triggered (z = 1) if the door is opened (y = 1) while it is still locked (x = 0) **for two consecutive clock cycles**. An example is shown below:

x	1	0	1	0	0	0	1	1	0	0	0	0	1	0
y	0	0	1	1	1	0	1	0	1	1	1	0	0	1
z	0	0	0	0	**1**	0	0	0	0	**1**	**1**	0	0	0

(a)	**Determine** the type of **FSM** that would be needed for the given problem and design the state diagram for the corresponding FSM.	[2]
(b)	**Derive** the state-assigned table using the **Gray encoding** from your state diagram in (a).	[2]
(c)	**Find** the expression for the next state and the output for the FSM using **KMAP**.	[3]
(d)	**Calculate** the number of flip-flops required for the circuit implementation and **draw** the circuit for this FSM.	[3]